

6th Gear Input / 1st Gear Output - Shaft Bending (Spur Gear)

$$W_t = \frac{2 \cdot T}{d} \quad W_v = W_t \cdot \tan(\phi_t)$$

$$T = 500 \text{ N}\cdot\text{m}$$

d = pitch diameter

$$d = 80 \text{ mm}$$

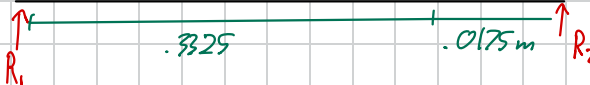
ϕ_t = Transverse pressure angle

$$W_{\text{total}} = 13248.807 \text{ N}$$

$$13248.807$$

$$W_t = 12500 \text{ N}$$

$$W_v = 4391 \text{ N}$$



$$\tau = \frac{T \cdot c}{J}$$

$$\tau = 318.3 \text{ MPa}$$

$$R_2 = 12586.366 \text{ N}$$

$$R_1 = 662.441 \text{ N}$$

$$M_R = 220.2616 \text{ N}\cdot\text{m}$$

$$\sigma = \frac{M \cdot c}{I} + \frac{W_a}{A} = \frac{220.2616 \cdot 0.01}{\frac{\pi}{64} \cdot (0.02)^4} + \frac{11491}{\pi \cdot (0.01)^2} = 280.446 \cdot 10^6 + 36.577 \cdot 10^6 = 317.023 \text{ MPa}$$

$$\sigma_{vm} = \sqrt{\sigma^2 + 3\tau^2}$$

$$\sigma_{vm} = 618.54 \text{ MPa}$$

6th Gear Input / 1st Gear Output - Shaft Bending

$$W_d = W_t \cdot \tan(\phi_H)$$

ϕ_H = Helix angle

$$\phi_H = 42.592$$

$$W_d = 11491 \text{ N}$$

$$\sigma_x = \sigma_b + \sigma_a = 569.9 \text{ MPa}$$

$$\sigma_{vm} = \sqrt{569.9^2 + 3(318.3)^2}$$

$$= 762.9 \text{ MPa}$$

$$\tau = 318.3 \text{ MPa}$$

$$\sigma_b = \frac{32 \cdot M}{\pi d^3} = \frac{32 \cdot 4188.44}{\pi \cdot (0.02)^3} = 533.3 \text{ MPa}$$

$$\sigma_a = \frac{W_d}{A} = \frac{11491}{\pi \cdot (0.01)^2} = 36.6 \text{ MPa}$$